

FREDRIK PETTIT

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Education

University of British Columbia

Vancouver, BC

Bachelor of Science in Computer Science, GPA: 3.8

Work Experience

ELLUSTROS

Jan 2026 – Present

Software Engineering Intern

The Blue Mountain, ON

- Architected a layered recommendation system separating frontend UI components from a Python service-layer engine, evaluating 15 ergonomic attributes across **100+ chairs** through structured seat-range and user preference models
- Designed a three-phase pipeline consisting of hard constraint filtering, weighted scoring, and ranked classification with explanation generation, incorporating deterministic tie-breaking, confidence scores, and structured missing-data handling
- Working within an Agile team, reduced recurring validation defects by **30%** across a multi-module B4A/B4X application by implementing 40+ targeted unit and regression tests covering boundary conditions and invalid inputs

UBC Department of Computer Science

Aug 2025 – Present

Teaching Assistant

Vancouver, BC

- Supported **900+** students in UBC's largest computer science course alongside 50+ TAs by leading tutorials and office hours
- Facilitated lab sessions for 30+ students in a team of 3, guiding systematic problem solving and conducting code reviews

Projects

YouTube Shorts AI Optimizer | *Python, FastAPI, AWS (Lambda, DynamoDB, SageMaker), OpenCV, XGBoost*

- Built an end-to-end video optimization pipeline automating thumbnail selection, title, and description generation for YouTube Shorts, growing a personal channel to **38,000+** subscribers, **45,000,000+** views, and **1,800,000+** likes
- Trained an XGBoost CTR prediction model on 3,000+ thumbnails via YouTube Data API, with features extracted from visual signals (HUD elements, action intensity, text overlays) using OpenCV and AWS Rekognition; deployed on SageMaker
- Designed an asynchronous serverless pipeline orchestrating video analysis, model inference, and content generation across distributed workers, with real-time job tracking, error recovery, and a FastAPI interface for status polling
- Integrated Gemini 1.5 Pro for video moment detection to identify high-engagement frames and Claude API for generating optimized titles and descriptions, reducing manual content optimization from hours to under **60** seconds per video

Inventory & Cashflow Manager | *Java, Spring Boot, React, PostgreSQL, Docker, Gradle*

- Engineered a multi-currency management system for a personal retail business tracking 50+ transactions and 40+ global shipments, processing **\$6K+** in cumulative cashflow with real-time currency conversion via the Frankfurter REST API
- Designed and exposed a RESTful API layer implementing full CRUD operations via Spring Data JPA and advanced analytical queries using JdbcTemplate against 10+ aggregated PostgreSQL views for profitability analysis
- Optimized query performance by **12×** compared to the initial implementation through indexing and query restructuring

Chelsea FC Play Predictor | *Python, pandas, scipy, PostgreSQL, Streamlit, scikit-learn*

- Built a pipeline ingesting Chelsea match data from FBref and Understat into PostgreSQL with automated feature computation and matchweek versioning, feeding 7 statistical and ML models producing probability distributions
- Engineered an EV optimizer scoring candidate picks against the app's partial-credit payoff function, validated through backtesting against baseline strategies; deployed as a Streamlit dashboard with picks, confidence, and EV breakdowns

Java OOP Quiz Application | *Java, Swing, JUnit, JSON*

- Developed a Java study tool for CPSC 210 students utilized by 5+ peers, implementing layered architecture with interface-based abstraction, encapsulated models, polymorphic JSON serialization, and a Singleton event logging system
- Built a JSON persistence layer for quiz state saving and loading, achieving 100% line and branch coverage with JUnit tests

Technical Skills

Languages: Java, Python, C++, C#, JavaScript, SQL, R, Racket

Frameworks & Libraries: Spring Boot, React, React Native, Expo, FastAPI, JUnit, Swing, scikit-learn, XGBoost, LightGBM, pandas, NumPy, scipy, OpenCV, Streamlit, BeautifulSoup, matplotlib, PyQt5

Cloud & DevOps: AWS (Lambda, SageMaker, DynamoDB, Step Functions, SQS, Rekognition), Docker, Gradle, Git

Databases & Tools: PostgreSQL, Firebase (Firestore & Storage), Jupyter Notebook, VS Code